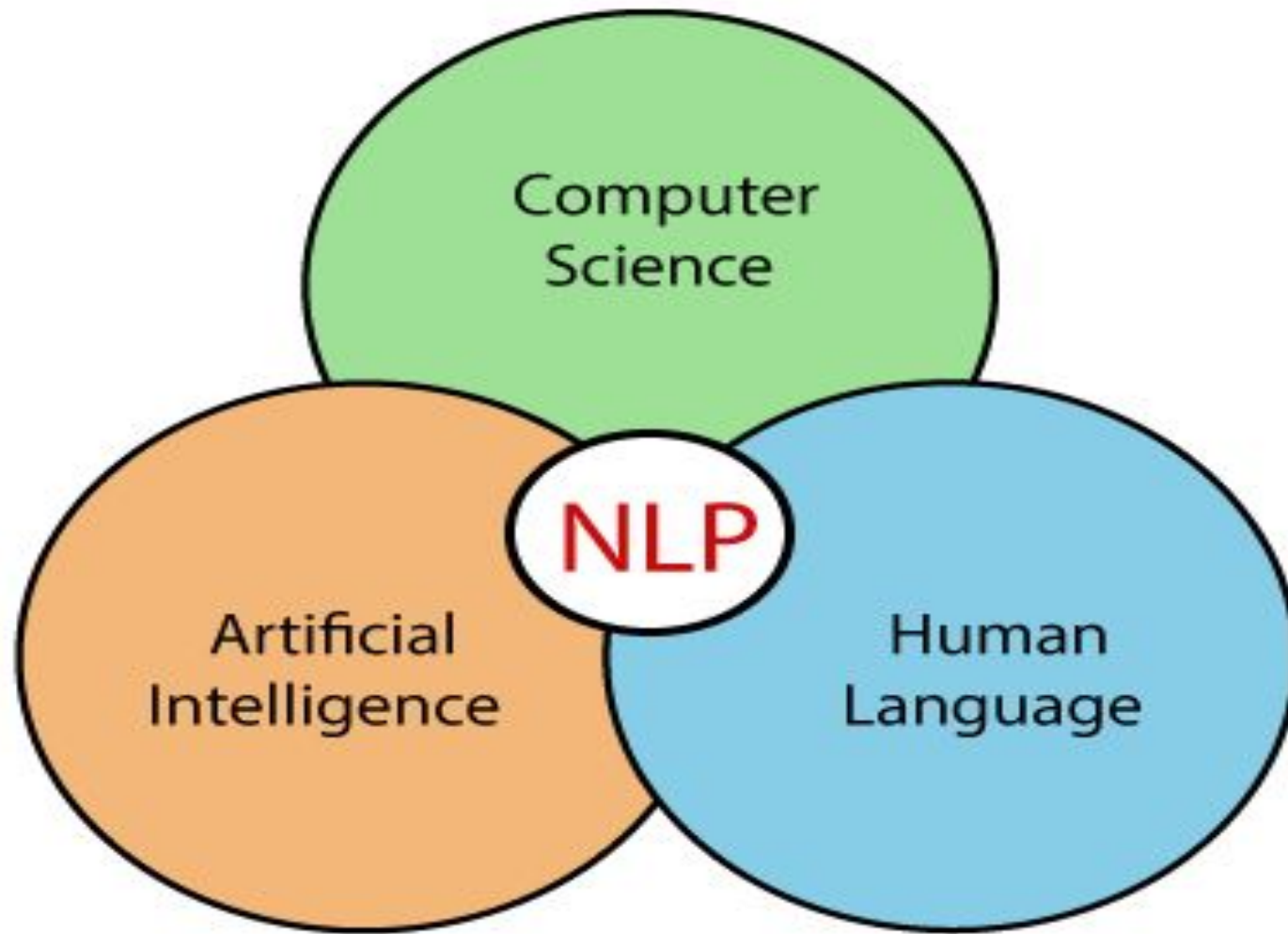


Module 4 : NLP

What Is Natural Language Processing?

- NLP stands for **Natural Language Processing**, which is a part of **Computer Science, Human language, and Artificial Intelligence**.
- It is the technology that is used by machines to understand, analyze, manipulate, and interpret human's languages. It helps developers to organize knowledge for performing tasks such as **translation, automatic summarization, Named Entity Recognition (NER), speech recognition, relationship extraction, and topic segmentation**.



<https://www.youtube.com/watch?v=6l-Alfkr5K4>

Components of NLP

- There are the following two components of NLP –
- **1. Natural Language Understanding (NLU)**
- Natural Language Understanding (NLU) helps the machine to understand and analyse human language by extracting the metadata from content such as concepts, entities, keywords, emotion, relations, and semantic roles.
- NLU mainly used in Business applications to understand the customer's problem in both spoken and written language.
- NLU involves the following tasks –
 - It is used to map the given input into useful representation.
 - It is used to analyze different aspects of the language.

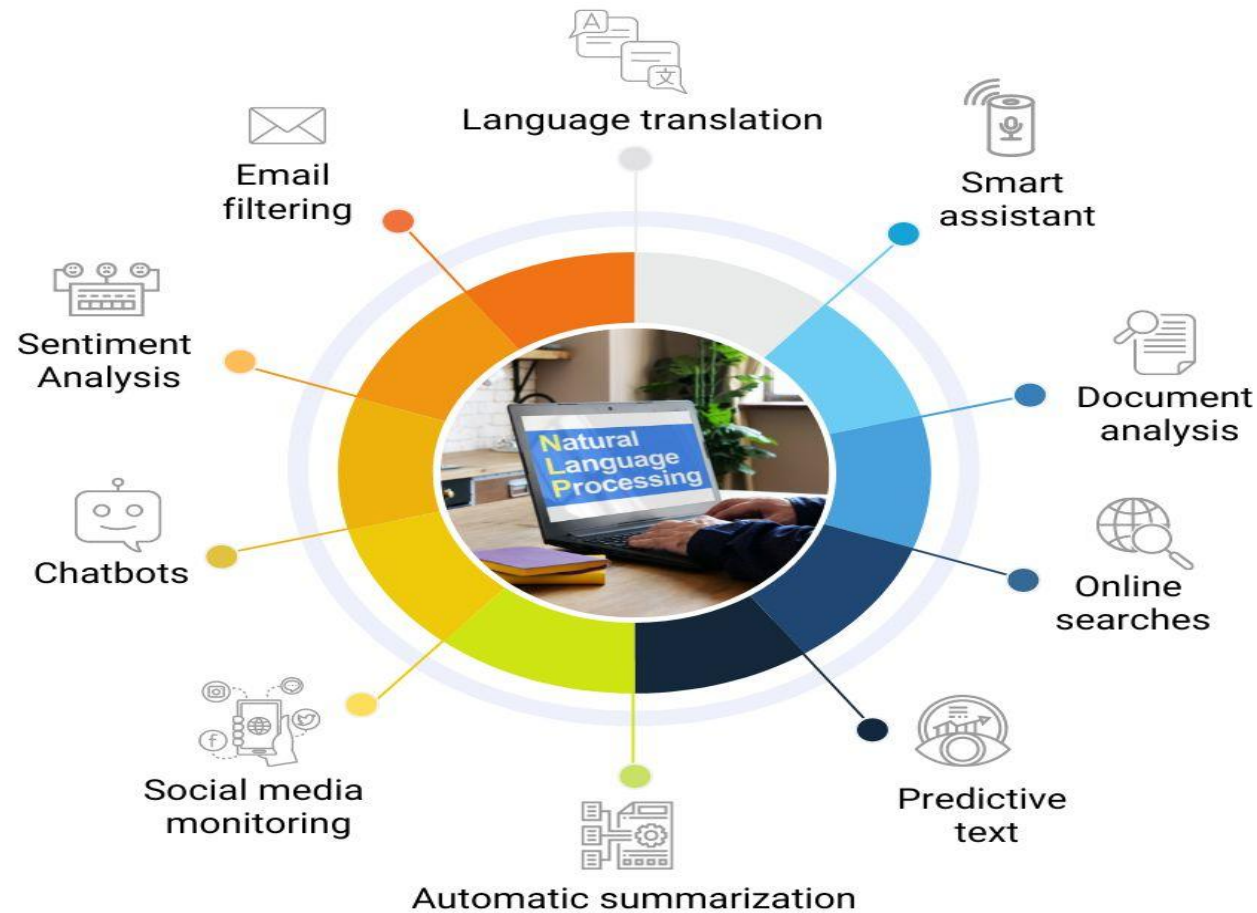
- **2. Natural Language Generation (NLG)**

- Natural Language Generation (NLG) acts as a translator that converts the computerized data into natural language representation. It mainly involves Text planning, Sentence planning, and Text Realization.
- Note: The NLU is difficult than NLG.

Difference between NLU and NLG

NLU	NLG
NLU is the process of reading and interpreting language.	NLG is the process of writing or generating language.
It produces non-linguistic outputs from natural language inputs.	It produces constructing natural language outputs from non-linguistic inputs.

Applications of Natural Language Processing



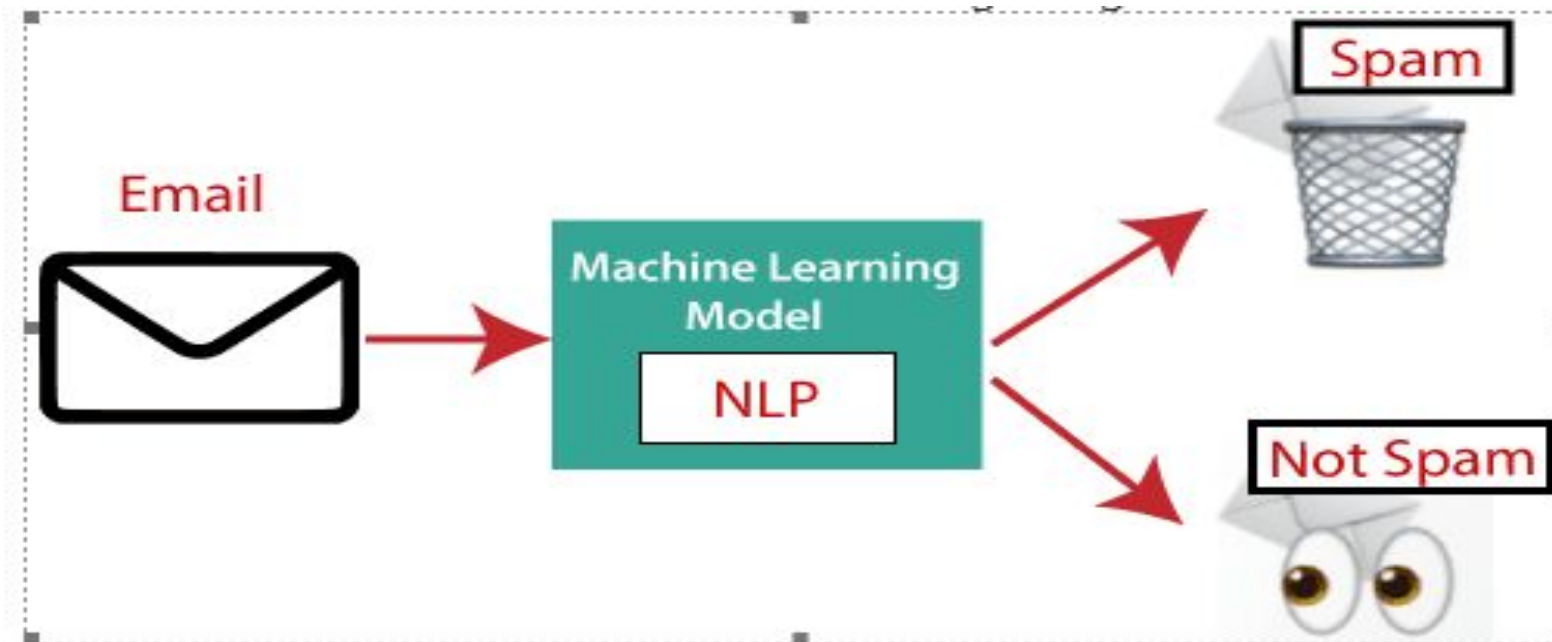
Applications of NLP

1.Question Answering Question Answering focuses on building systems that automatically answer the questions asked by humans in a natural language.



- **2. Spam Detection**

- Spam detection is used to detect unwanted e-mails getting to a user's inbox.



- **3. Sentiment Analysis**

- Sentiment Analysis is also known as **opinion mining**. It is used on the web to analyze the attitude, behavior, and emotional state of the sender. This application is implemented through a combination of NLP (Natural Language Processing) and statistics by assigning the values to the text (positive, negative, or neutral), identify the mood of the context (happy, sad, angry, etc.)



4. Machine Translation

- Machine translation is used to translate text or speech from one natural language to another natural language.
- **Example:** Google Translator

- 5. Spelling correction
- Microsoft Corporation provides word processor software like MS-word, PowerPoint for the spelling correction.



6. Speech Recognition

- Speech recognition is used for converting spoken words into text. It is used in applications, such as mobile, home automation, video recovery, dictating to Microsoft Word, voice biometrics, voice user interface, and so on.

7. Chatbot

- Implementing the Chatbot is one of the important applications of NLP. It is used by many companies to provide the customer's chat services.

- **8. Information extraction**

- Information extraction is one of the most important applications of NLP. It is used for extracting structured information from unstructured or semi-structured machine-readable documents.

- **9. Natural Language Understanding (NLU)**

- It converts a large set of text into more formal representations such as first-order logic structures that are easier for the computer programs to manipulate notations of the natural language processing.

How to build an NLP pipeline

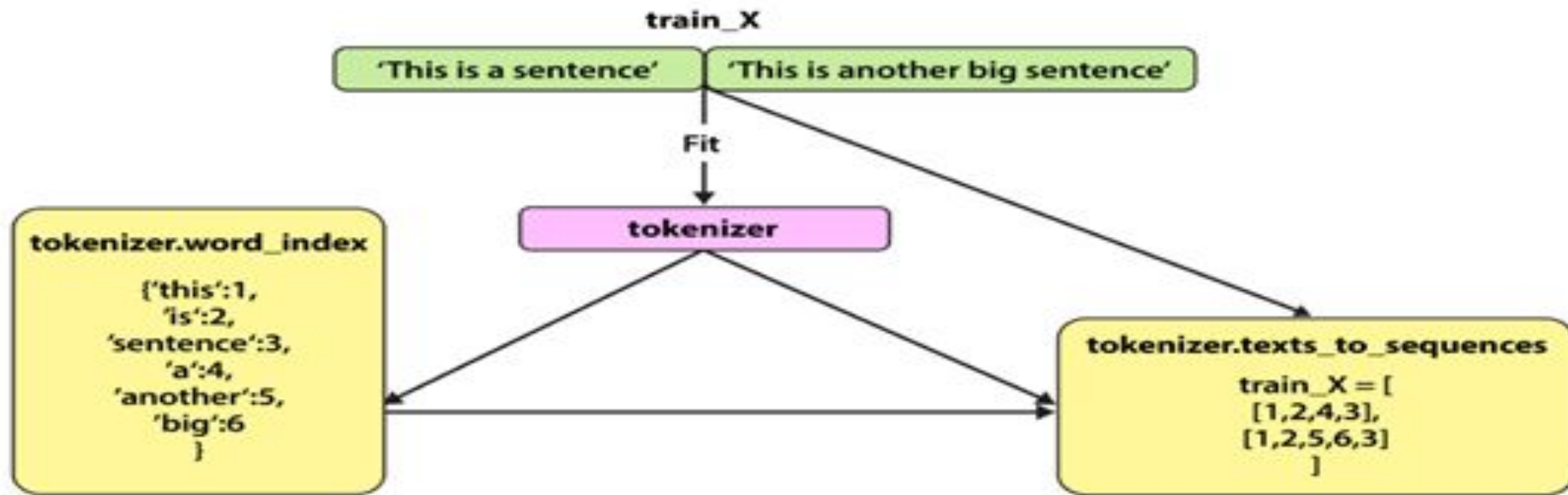
- Step1: Sentence Segmentation
- Step2: Word Tokenization
- Step3: Stemming
- Step 4: Lemmatization
- Step 5: Identifying Stop Words
- Step 6: Dependency Parsing
- Step 7: POS tags
- Step 8: Named Entity Recognition (NER)
- Step 9: Chunking

- **Step1: Sentence Segmentation**

- Sentence Segment is the first step for building the NLP pipeline. It breaks the paragraph into separate sentences.
- **Example:** Consider the following paragraph –
- **Independence Day is one of the important festivals for every Indian citizen. It is celebrated on the 15th of August each year ever since India got independence from the British rule. The day celebrates independence in the true sense.**
- **Sentence Segment produces the following result:**
 1. "Independence Day is one of the important festivals for every Indian citizen."
 2. "It is celebrated on the 15th of August each year ever since India got independence from the British rule."
 3. "This day celebrates independence in the true sense."

- Step2: Word Tokenization

TOKENIZERS



Given a corpus of documents, a tokenizer maps every word to an index. Then it can translate any document into a sequence of numbers.

- Word Tokenizer is used to break the sentence into separate words or tokens.
- **Example:**
- JavaTpoint offers Corporate Training, Summer Training, Online Training, and Winter Training.
- Word Tokenizer generates the following result:
- "JavaTpoint", "offers", "Corporate", "Training", "Summer", "Training", "Online", "Training", "and", "Winter", "Training", ""

- **Step3: Stemming**

- Stemming is used to normalize words into its base form or root form. For example, celebrates, celebrated and celebrating, all these words are originated with a single root word "celebrate." The big problem with stemming is that sometimes it produces the root word which may not have any meaning.
- **For Example**, intelligence, intelligent, and intelligently, all these words are originated with a single root word "intelligen." In English, the word "intelligen" do not have any meaning.

- **Step 4: Lemmatization**

- Lemmatization is quite similar to the Stemming. It is used to group different inflected forms of the word, called Lemma. The main difference between Stemming and lemmatization is that it produces the root word, which has a meaning.
- **For example:** In lemmatization, the words intelligence, intelligent, and intelligently has a root word intelligent, which has a meaning.
-

- **Step 5: Identifying Stop Words**

- In English, there are a lot of words that appear very frequently like "is", "and", "the", and "a". NLP pipelines will flag these words as stop words. **Stop words** might be filtered out before doing any statistical analysis.
- **Example:** He is a good boy.
- **Note:** When you are building a rock band search engine, then you do not ignore the word "The."

- **Step 6: Dependency Parsing**

- Dependency Parsing is used to find that how all the words in the sentence are related to each other.
-

- **Step 7: POS tags**

- POS stands for parts of speech, which includes Noun, verb, adverb, and Adjective. It indicates that how a word functions with its meaning as well as grammatically within the sentences. A word has one or more parts of speech based on the context in which it is used.
- **Example: "Google"** something on the Internet.
- In the above example, Google is used as a verb, although it is a proper noun.

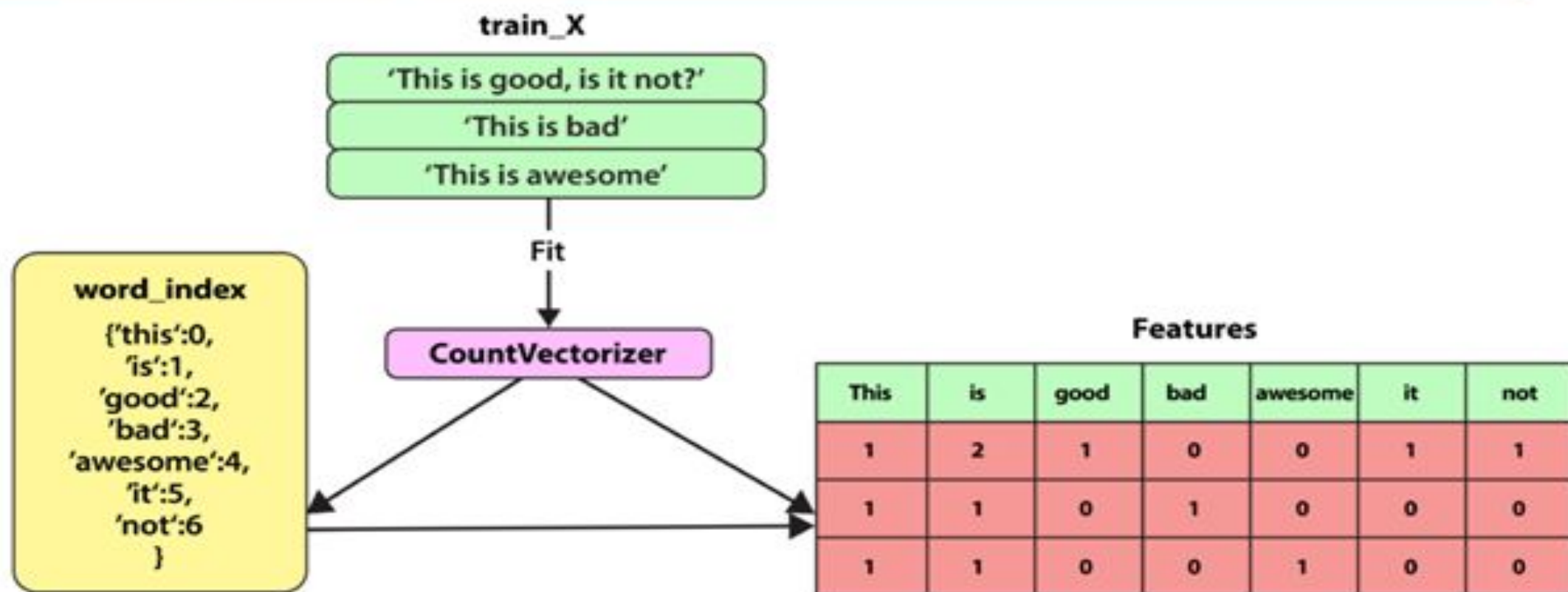
- Step 8: Named Entity Recognition (NER)

NAMED ENTITY RECOGNITION (NER) TAGGING

Andrew Yan-Tak Ng **PERSON** (**Chinese** **NORP** : 吳恩達; born **1976** **DATE**) is a **British** **NORP** -born **American** **NORP** computer scientist and technology entrepreneur focusing on machine learning and **AI** **GPE** .
Ng was a co-founder and head of **Google Brain** **ORG** and was the former chief scientist at **Baidu** **ORG** ,
building the company's **Artificial Intelligence Group** **ORG** into a team of **several thousand** **CARDINAL** people.

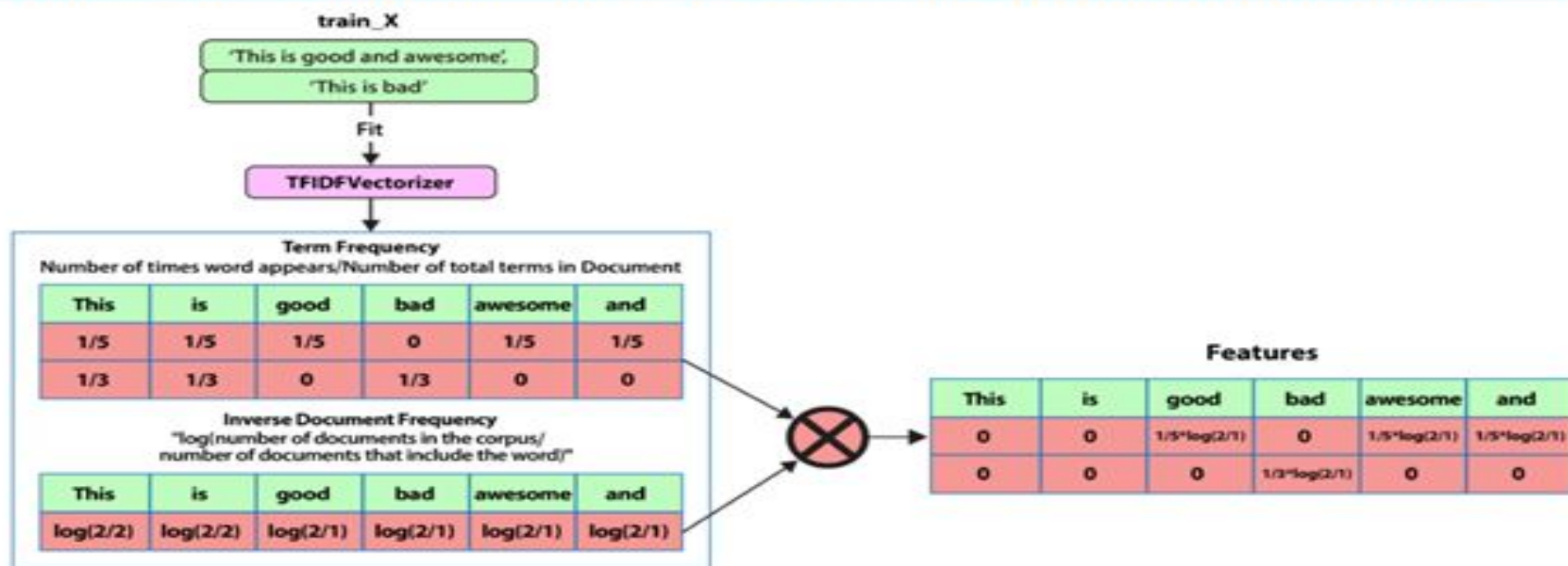
spaCy named entity recognition tagging of the first paragraph of Andrew Ng's Wikipedia page. "NORP" stands for nationalities or religious or political groups. Note that spaCy incorrectly labels "AI" as "GPE," for geopolitical entity.

TOKENIZERS: BAG-OF-WORDS



Bag-of-Words (through the CountVectorizer method) encodes the total number of times a document uses each word in the associated corpus.

TOKENIZERS: TERM FREQUENCY - INVERSE DOCUMENT FREQUENCY (TF-IDF)



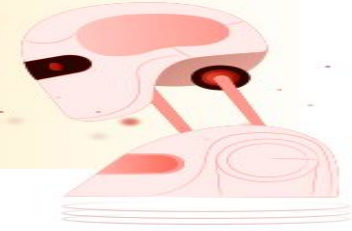
TF-IDF creates features for each document based on how often each word shows up in a document versus the entire corpus.

- Named Entity Recognition (NER) is the process of detecting the named entity such as person name, movie name, organization name, or location.
- **Example: Steve Jobs** introduced iPhone at the Macworld Conference in San Francisco, California.

- **Step 9: Chunking**

- Chunking is used to collect the individual piece of information and grouping them into bigger pieces of sentences.

Top 14 Use Cases of NLP in Healthcare



Clinical Documentation



Speech Recognition



Computer-assisted Coding



Data Mining Research



Automated Registry Reporting



Clinical Decision Support



Clinical Trial Matching



Prior Authorization



AI Chatbots and Virtual Scribe



Risk Adjustment Model



Computational Phenotyping



Review Management & Sentiment Analysis



Dictation and EMR Implications



Root Cause Analysis

Advantages of NLP

- NLP helps users to ask questions about any subject and get a direct response within seconds.
- NLP offers exact answers to the question means it does not offer unnecessary and unwanted information.
- NLP helps computers to communicate with humans in their languages.
- It is very time efficient.
- Most of the companies use NLP to improve the efficiency of documentation processes, accuracy of documentation, and identify the information from large databases.

Disadvantages of NLP

- NLP may not show context.
- NLP is unpredictable
- NLP may require more keystrokes.
- NLP is unable to adapt to the new domain, and it has a limited function that's why NLP is built for a single and specific task only.

Phases of NLP



1. Lexical Analysis and Morphological

- The first phase of NLP is the Lexical Analysis. This phase scans the source code as a stream of characters and converts it into meaningful lexemes. It divides the whole text into paragraphs, sentences, and words.

2. Syntactic Analysis (Parsing)

- Syntactic Analysis is used to check grammar, word arrangements, and shows the relationship among the words.
- **Example:** Agra goes to the Poonam
- In the real world, Agra goes to the Poonam, does not make any sense, so this sentence is rejected by the Syntactic analyzer.

3. Semantic Analysis

- Semantic analysis is concerned with the meaning representation. It mainly focuses on the literal meaning of words, phrases, and sentences.

4. Discourse Integration

- Discourse Integration depends upon the sentences that precedes it and also invokes the meaning of the sentences that follow it.

5. Pragmatic Analysis

- Pragmatic is the fifth and last phase of NLP. It helps you to discover the intended effect by applying a set of rules that characterize cooperative dialogues.
- **For Example:** "Open the door" is interpreted as a request instead of an order.

Why NLP is difficult?

- NLP is difficult because Ambiguity and Uncertainty exist in the language.
- **Ambiguity**
- There are the following three ambiguity –
 - **Lexical Ambiguity**
- Lexical Ambiguity exists in the presence of two or more possible meanings of the sentence within a single word.
- **Example:** Manya is looking for a **match**.
- In the above example, the word match refers to that either Manya is looking for a partner or Manya is looking for a match. (Cricket or other match)

- **Syntactic Ambiguity**

- Syntactic Ambiguity exists in the presence of two or more possible meanings within the sentence.
- **Example:**
- I saw the girl with the binocular.
- In the above example, did I have the binoculars? Or did the girl have the binoculars?

- **Referential Ambiguity**

- Referential Ambiguity exists when you are referring to something using the pronoun.
- **Example:** Kiran went to Sunita. She said, "I am hungry."
- In the above sentence, you do not know that who is hungry, either Kiran or Sunita.

NLP APIs

- Natural Language Processing APIs allow developers to integrate human-to-machine communications and complete several useful tasks such as speech recognition, chatbots, spelling correction, sentiment analysis, etc.

IBM Watson API

- IBM Watson API combines different sophisticated machine learning techniques to enable developers to classify text into various custom categories.
- It supports multiple languages, such as English, French, Spanish, German, Chinese, etc.
- With the help of IBM Watson API, you can extract insights from texts, add automation in workflows, enhance search, and understand the sentiment.
- The main advantage of this API is that it is very easy to use.
- **Pricing:** Firstly, it offers a free 30 days trial IBM cloud account. You can also opt for its paid plans.

Chatbot API

- Chatbot API allows you to create intelligent chatbots for any service.
- It supports Unicode characters, classifies text, multiple languages, etc.
- It is very easy to use.
- It helps you to create a chatbot for your web applications.

Pricing: Chatbot API is free for 150 requests per month. You can also opt for its paid version, which starts from \$100 to \$5,000 per month.

Speech to text API

- Speech to text API is used to convert speech to text

Pricing: Speech to text API is free for converting 60 minutes per month. Its paid version starts from \$500 to \$1,500 per month.

Sentiment Analysis API

- Sentiment Analysis API is also called as '**opinion mining**' which is used to identify the tone of a user (positive, negative, or neutral)

Pricing: Sentiment Analysis API is free for less than 500 requests per month. Its paid version starts from \$19 to \$99 per month.

Translation API by SYSTRAN

- The Translation API by SYSTRAN is used to translate the text from the source language to the target language.
- You can use its NLP APIs for language detection, text segmentation, named entity recognition, tokenization, and many other tasks.

Pricing: This API is available for free. But for commercial users, you need to use its paid version.

Text Analysis API by AYLIEN

- Text Analysis API by AYLIEN is used to derive meaning and insights from the textual content.
- It is available for both free as well as paid from \$119 per month. It is easy to use.

Pricing: This API is available free for 1,000 hits per day. You can also use its paid version, which starts from \$199 to \$1,399 per month.

Cloud NLP API

- The Cloud NLP API is used to improve the capabilities of the application using natural language processing technology.
- It allows you to carry various natural language processing functions like sentiment analysis and language detection.
- It is easy to use.

Pricing: Cloud NLP API is available for free.

Google Cloud Natural Language API

- Google Cloud Natural Language API allows you to extract beneficial insights from unstructured text.
- This API allows you to perform entity recognition, sentiment analysis, content classification, and syntax analysis in more the 700 predefined categories.
- It also allows you to perform text analysis in multiple languages such as English, French, Chinese, and German.

Pricing: After performing entity analysis for 5,000 to 10,000,000 units, you need to pay \$1.00 per 1000 units per month.

NLP Libraries

- **Scikit-learn:** It provides a wide range of algorithms for building machine learning models in Python.
- **Natural language Toolkit (NLTK):** NLTK is a complete toolkit for all NLP techniques.
- **Pattern:** It is a web mining module for NLP and machine learning.
- **TextBlob:** It provides an easy interface to learn basic NLP tasks like sentiment analysis, noun phrase extraction, or pos-tagging.
- **Quepy:** Quepy is used to transform natural language questions into queries in a database query language.
- **SpaCy:** SpaCy is an open-source NLP library which is used for Data Extraction, Data Analysis, Sentiment Analysis, and Text Summarization.
- **Gensim:** Gensim works with large datasets and processes data streams.

Difference between Natural language and Computer Language

	Natural Language	Computer Language
vocabulary	a very large vocabulary.	a very limited vocabulary.
Understanding of language	easily understood by humans.	easily understood by the machines.
Ambiguity	ambiguous in nature	unambiguous

Six Important Natural Language Processing (NLP) Models

- **Eliza**

- developed in the mid-1960s to try to solve the Turing Test
- to fool people into thinking they're conversing with another human being rather than a machine.
- Eliza used pattern matching and a series of rules without encoding the context of the language.

- **Tay**

- chatbot that Microsoft launched in 2016.
- It was supposed to tweet like a [teen](#) and learn from conversations with real users on Twitter.
- The bot adopted phrases from users who tweeted sexist and racist comments, and Microsoft deactivated it not long afterward.
- Tay illustrates some points made by the “Stochastic Parrots” paper, particularly the danger of not debiasing data.

- **BERT** and his Muppet friends:

- Many deep learning models for NLP are [named after Muppet characters](#), including [ELMo](#), [BERT](#), [Big BIRD](#), [ERNIE](#), [Kermit](#), [Grover](#), [RoBERTa](#), and [Rosita](#).
- Most of these models are good at providing contextual embeddings and enhanced knowledge representation.

- **Generative Pre-Trained Transformer 3 (GPT-3)**
- 175 billion parameter model that can write original prose with [human-equivalent fluency](#) in response to an input prompt.
- The model is based on the transformer architecture. The previous version, GPT-2, is open source.
- [Microsoft acquired an exclusive license](#) to access GPT-3's underlying model from its developer OpenAI, but other users can interact with it via an application programming interface (API).
- Several groups including [EleutherAI](#) and [Meta](#) have released open source interpretations of GPT-3.

- **Language Model for Dialogue Applications (LaMDA)**
- conversational chatbot developed by Google.
- LaMDA is a transformer-based model trained on dialogue rather than the usual web text.
- The system aims to provide sensible and specific responses to conversations.
- Google developer Blake Lemoine came to believe that LaMDA is sentient.
- Lemoine had detailed conversations with AI about his rights and personhood.
- During one of these conversations, the AI changed Lemoine's mind about Isaac Asimov's third law of robotics.
- Lemoine claimed that LaMDA was sentient, but the idea was disputed by many observers and commentators.
- Subsequently, Google placed Lemoine on administrative leave for distributing proprietary information and ultimately fired him.

- **Mixture of Experts (MoE):**

- While most deep learning models use the same set of parameters to process every input, MoE models aim to provide different parameters for different inputs based on efficient routing algorithms to achieve [higher performance](#).
- [Switch Transformer](#) is an example of the MoE approach that aims to reduce communication and computational costs.

NLP in medicine

1. Improving Patient Care

- Primary benefits of NLP in healthcare is its ability to improve patient care.
- Healthcare professionals can use NLP to extract relevant information from patient records, such as medical history, medication allergies, and previous diagnoses.
- This information can be used to develop personalized treatment plans for patients.
- NLP can also help identify patients who are at high risk of developing certain conditions, allowing healthcare professionals to intervene early and prevent the development of the disease.

NLP in medicine

2. Enhancing Medical Research

- NLP can also be used to analyze vast amounts of medical data to identify patterns and trends.
- This can help researchers develop new treatments and therapies.
- For example, NLP can be used to analyze patient data to determine which treatments are most effective for certain conditions.
- It can also help identify the side effects of different medications, allowing researchers to develop safer and more effective treatments.

3. Improving Clinical Trials

- Used to improve clinical trials by making the recruitment process more efficient.
- Clinical trials require a large number of participants and finding suitable candidates can be time-consuming and expensive.
- NLP can be used to analyze patient data to identify suitable candidates for clinical trials, reducing the time and cost required to recruit participants.
- Improving the recruitment process is just one of the ways NLP can benefit clinical trials.
- By analyzing patient data, NLP can help identify patients who meet the specific inclusion criteria for a clinical trial.
- This process can be time-consuming and labor-intensive if done manually, but NLP can speed up the process significantly.
- NLP algorithms can sift through a large amount of data and extract information relevant to the clinical trial.
- This information can include medical history, previous diagnoses, medication usage, and other factors that might make a patient suitable for a particular trial.
- By automating this process, researchers can save time and money while increasing the likelihood of finding suitable participants.

4. Enhancing Electronic Health Records (EHRs)

- NLP can also be used to improve the accuracy and completeness of electronic health records (EHRs).
- EHRs are digital versions of patient medical records that contain information about a patient's medical history, diagnosis, and treatment plan.
- NLP can help healthcare professionals extract relevant information from these records, ensuring that they are accurate and up-to-date.
- This can help improve patient care by providing healthcare professionals with the information they need to make informed decisions about a patient's treatment plan.

5. Assisting Healthcare Professionals

- NLP can also be used to assist healthcare professionals in their day-to-day tasks.
- For example, it can be used to transcribe physician notes, allowing them to focus on patient care instead of documentation.
- It can also be used to identify potential drug interactions and side effects, allowing healthcare professionals to adjust a patient's treatment plan accordingly.
- NLP has the potential to assist healthcare professionals in a wide range of day-to-day tasks.

- **Transcribing physician notes:**

- NLP can be used to transcribe physician notes, which is a time-consuming and often error-prone task.
- By using NLP to transcribe notes, healthcare professionals can save time and reduce errors, allowing them to focus on providing patient care instead of documentation.

- **Extracting information from medical literature:**

- NLP can be used to extract information from medical literature, making it easier for healthcare professionals to stay up-to-date on the latest research and best practices

Low-Level Components

1. Tokenization:

- Tokenization is the process of breaking down a text into individual words or tokens.
- These tokens are the basic units of text used for further analysis.
- For example, the sentence "I love ice cream" would be tokenized into ["I", "love", "ice", "cream"].

2.Part-of-Speech Tagging (POS):

- POS tagging involves assigning grammatical labels (such as noun, verb, adjective, etc.) to each token in a sentence.
- For example, in the sentence "He runs fast," POS tagging would label "He" as a pronoun, "runs" as a verb, and "fast" as an adverb.

3. Named Entity Recognition (NER):

- NER identifies and classifies entities in text, such as names of people, organizations, locations, dates, and more.
- For example, in the sentence "Apple Inc. is headquartered in Cupertino," NER would recognize "Apple Inc." as an organization and "Cupertino" as a location.

4. Stemming and Lemmatization:

- These processes reduce words to their base or root forms. Stemming removes prefixes and suffixes, while lemmatization considers the word's meaning to find its base form.
- For example, "running" would be stemmed to "run" and lemmatized to "run."

High-Level Components

1. Syntactic Analysis (Parsing):

- Syntactic analysis involves determining the grammatical structure and relationships between words in a sentence.
- It helps in understanding sentence structure and dependencies. Parsing often produces tree structures like parse trees or dependency trees.

2. Semantic Analysis:

- Semantic analysis focuses on understanding the meaning of text.
- It can involve tasks like semantic role labeling (identifying who does what to whom in a sentence) and semantic similarity (measuring how similar the meanings of two sentences are).

High-Level Components

3.Text Classification:

- Text classification assigns predefined categories or labels to documents or sentences.
- It's used in tasks like sentiment analysis, spam detection, and topic categorization.

4. Machine Translation:

- Machine translation is a high-level NLP task that involves translating text from one language to another.
- It requires understanding the meaning of the source language and generating equivalent text in the target language.

- **5.Question Answering (QA):**

- QA systems take a question in natural language and attempt to provide a relevant answer by processing a large corpus of text or knowledge bases.

6. Text Generation:

- Text generation tasks involve creating human-like text, such as generating summaries, writing coherent paragraphs, or even composing poetry.
- This often requires a deep understanding of language and context.

7. Conversational Agents (Chatbots):

- Building conversational agents that can engage in natural language conversations with users involves high-level NLP tasks like dialogue management, intent recognition, and response generation.

- Low-level components typically provide the foundational processing needed to prepare text data for high-level analysis and understanding.
- High-level components focus on tasks that require a deeper understanding of language and context, often involving complex reasoning and decision-making processes.
- Together, these components enable the development of sophisticated NLP applications and systems.

